





ICTAI 2026

The 38th IEEE International Conference on Tools with Artificial Intelligence

In-Context Learning for Foundation Time-series Models with Prior-Data Fitted Networks

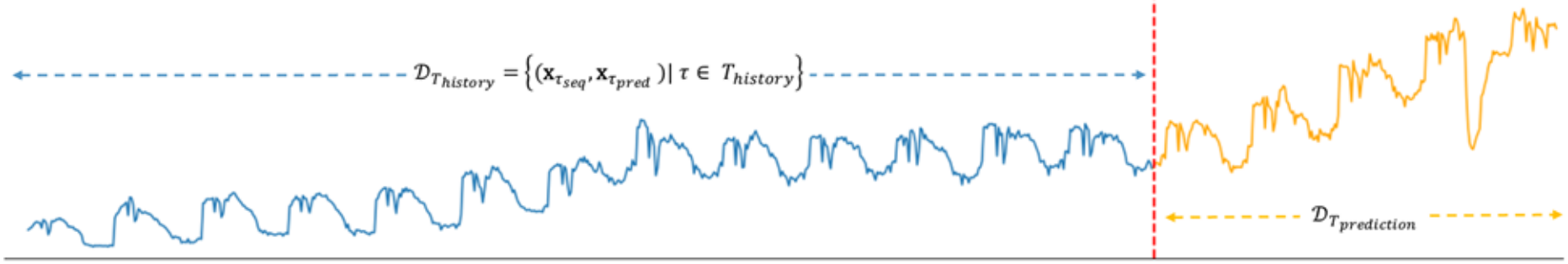
2-4 November 2026 – Boca Raton FL, USA

Long Vu, Joseph Natter, Madeline Frank,
Bobin Deng, Rui Wu, Xiaofeng Wu, Kun Suo

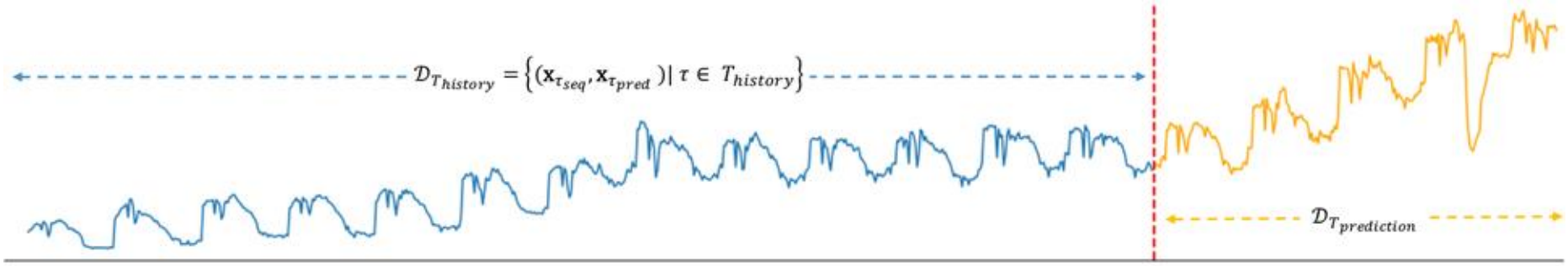
Outline

1. Motivation
    a. Time-series Forecasting
    b. In-context Learning
2. Methodology
3. Results
4. Conclusion

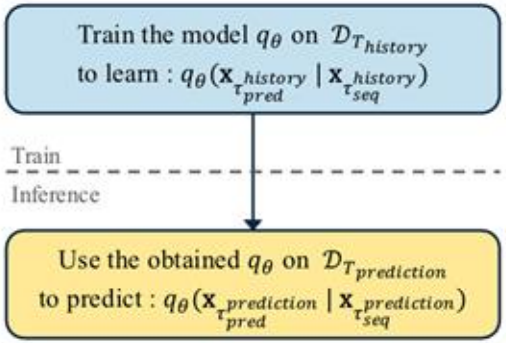
Time-series Forecasting



(a) Example of a target real-world time-series: $\mathcal{D} \equiv (\mathcal{D}_{T_{history}}, \mathcal{D}_{T_{prediction}})$

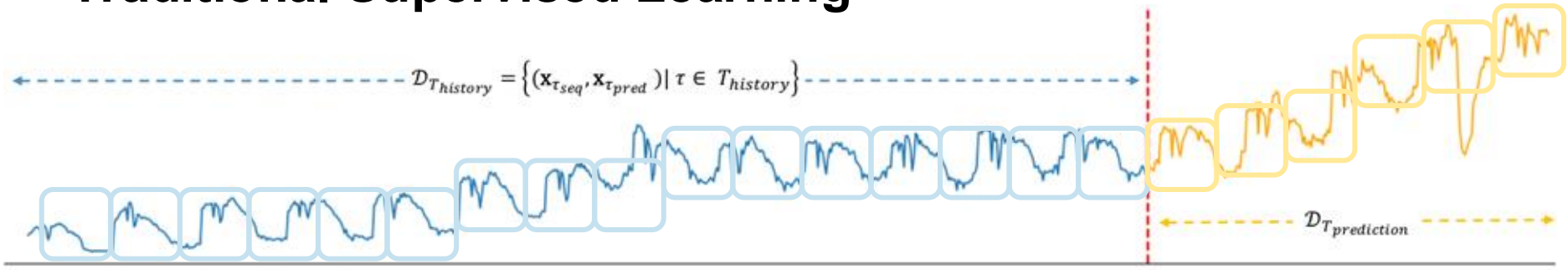


(a) Example of a target real-world time-series: $\mathcal{D} \equiv (\mathcal{D}_{T_{history}}, \mathcal{D}_{T_{prediction}})$

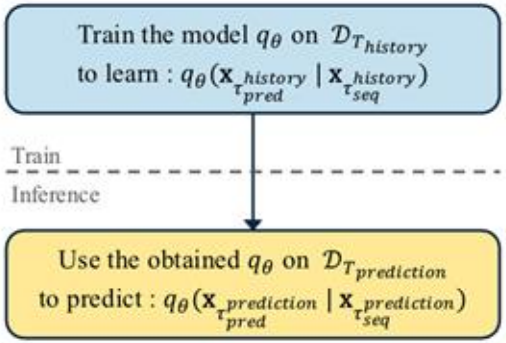


(b) Traditional Models

Traditional Supervised Learning

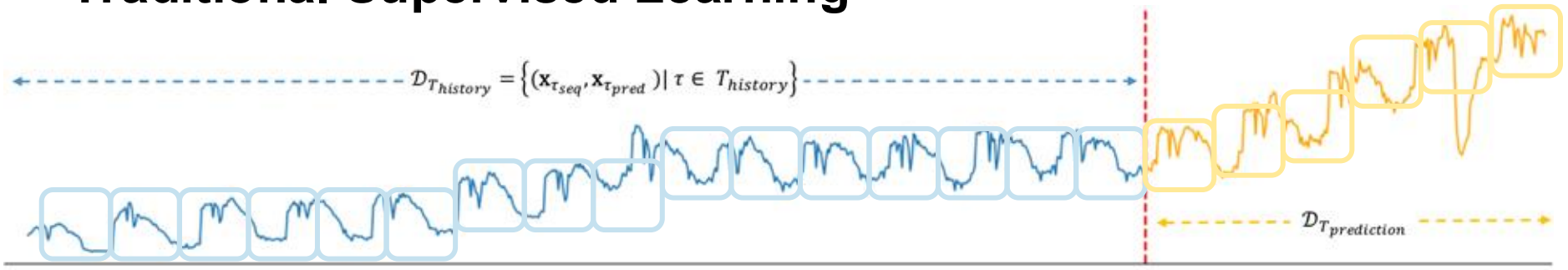


(a) Example of a target real-world time-series: $\mathcal{D} \equiv (\mathcal{D}_{T_{history}}, \mathcal{D}_{T_{prediction}})$

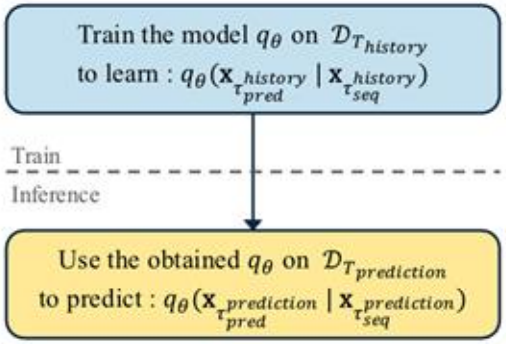


(b) Traditional Models

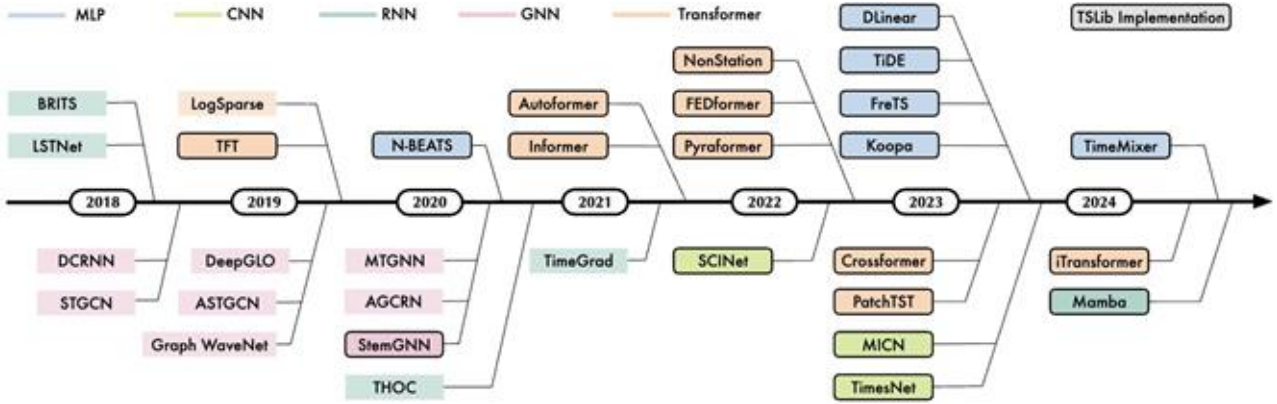
Traditional Supervised Learning



(a) Example of a target real-world time-series: $\mathcal{D} \equiv (\mathcal{D}_{T_{history}}, \mathcal{D}_{T_{prediction}})$



(b) Traditional Models



Wang, Y., Wu, H., Dong, J., Liu, Y., Wang, C., Long, M., & Wang, J. (2024). Deep time series models: A comprehensive survey and benchmark. arXiv preprint arXiv:2407.13278.

Zero-Shot Learning

- Conceptually: pretrain a model once on an external corpus.
- Then, evaluate it on new datasets with no training or parameter updates

Zero-Shot Learning

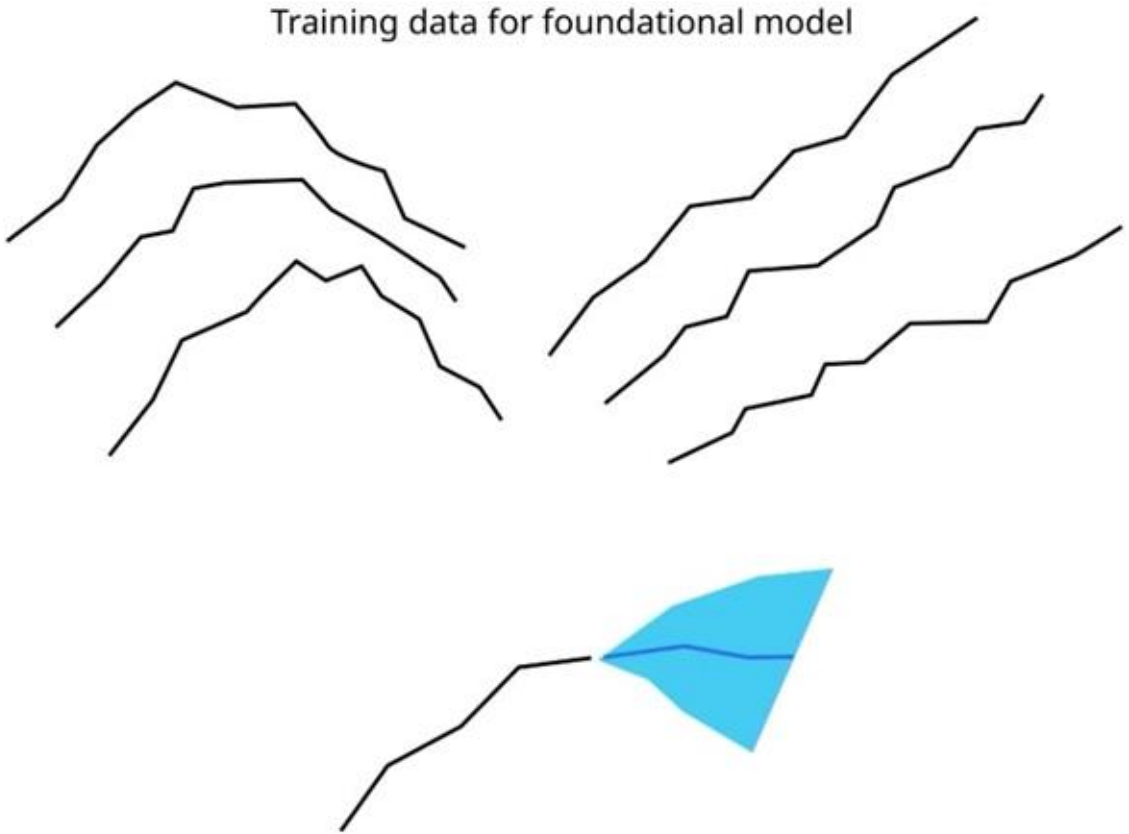
- Conceptually: pretrain a model once on an external corpus.
- Then, evaluate it on new datasets with no training or parameter updates

Tan, M., Merrill, M., Gupta, V., Althoff, T., & Hartvigsen, T. (2024). Are language models actually useful for time series forecasting?. Advances in Neural Information Processing Systems, 37, 60162-60191.

C. Bergmeir, “LLMs and foundational models: Not (yet) as good as hoped,” Foresight: The International Journal of Applied Forecasting, vol. 73, 2024.

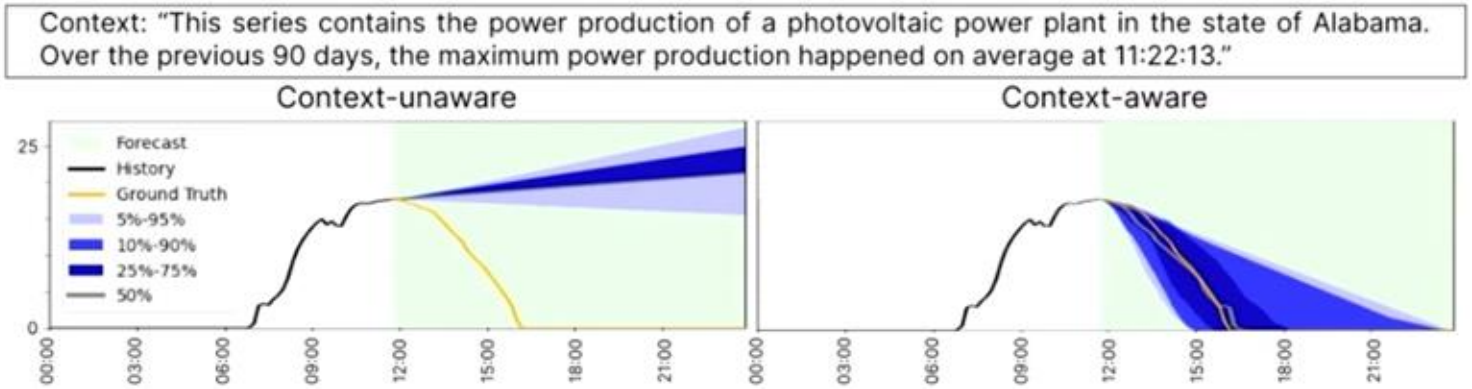
Drawbacks:

- uncertainty explosion
- poor inductive bias
- brittle extrapolation



The Missing Ingredient: Context in Inference

- More data ≠ better model; what we need is contextual inference
- Time-series naturally provide historical context



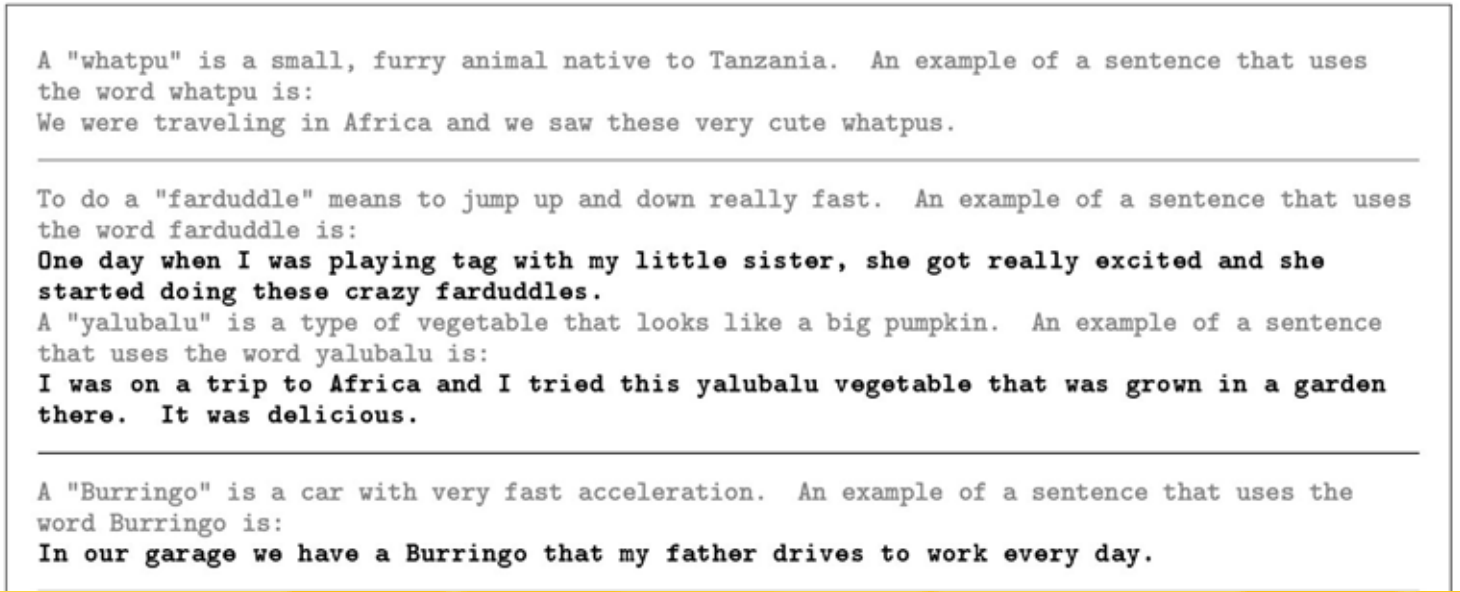
Source: Williams, A. R., Ashok, A., Marcotte, É., Zantedeschi, V., Subramanian, J., Riachi, R., ... & Drouin, A. (2024). Context is Key: A Benchmark for Forecasting with Essential Textual Information. arXiv preprint arXiv:2410.18959.

In-context Learning

In-Context Learning

Radford et al., “Language Models are Unsupervised Multitask Learners,” OpenAI, 2019.

Brown et al., “Language Models are Few-Shot Learners,” NeurIPS, 2020. (image credit)



Prior-data Fitted Network (intuition)

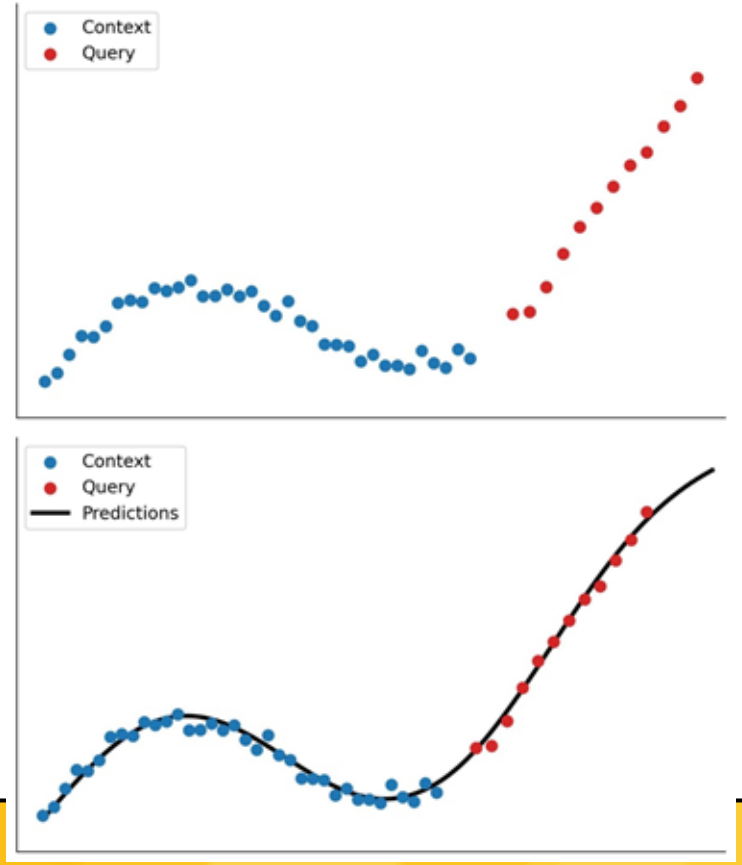
Leverage In-Context Learning with the pipeline:

- Generate data that represents prior beliefs (families of generating functions)
- Repeatedly sample from this data to train a model
- At evaluation, show the model context and query. The model will infer which function generated the context, and use it to extrapolate from the query.

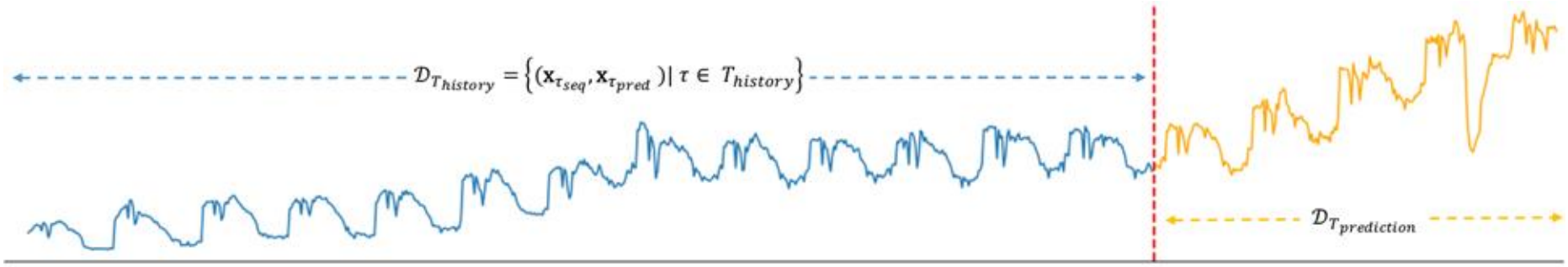
```
To do a "farduddle" means to jump up and down really fast.  An example of a sentence that uses
the word farduddle is:
One day when I was playing tag with my little sister, she got really excited and she
started doing these crazy farduddles.
```

Context, query, **answer**

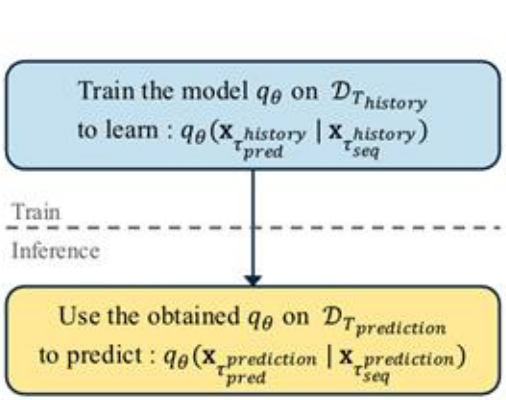
Prior-data Fitted Network (intuition)



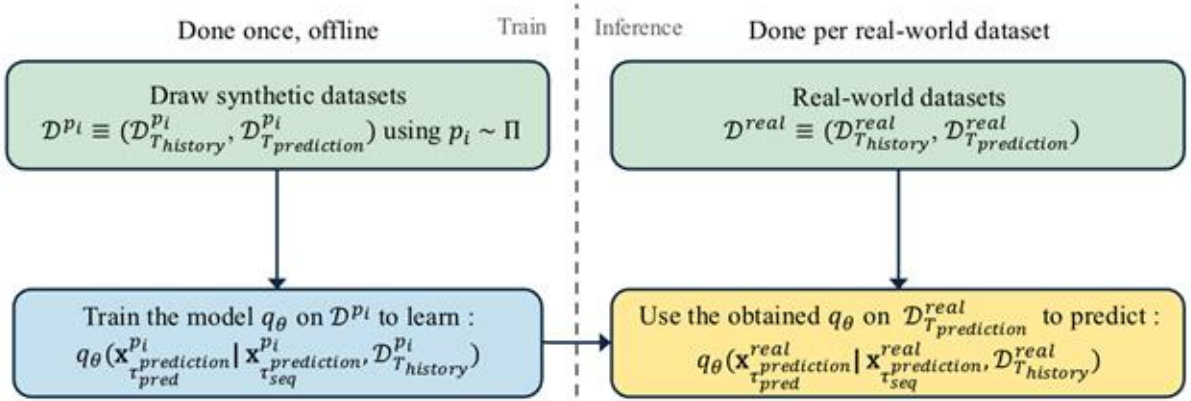
Methodology



(a) Example of a target real-world time-series: $\mathcal{D} \equiv (\mathcal{D}_{T_{history}}, \mathcal{D}_{T_{prediction}})$

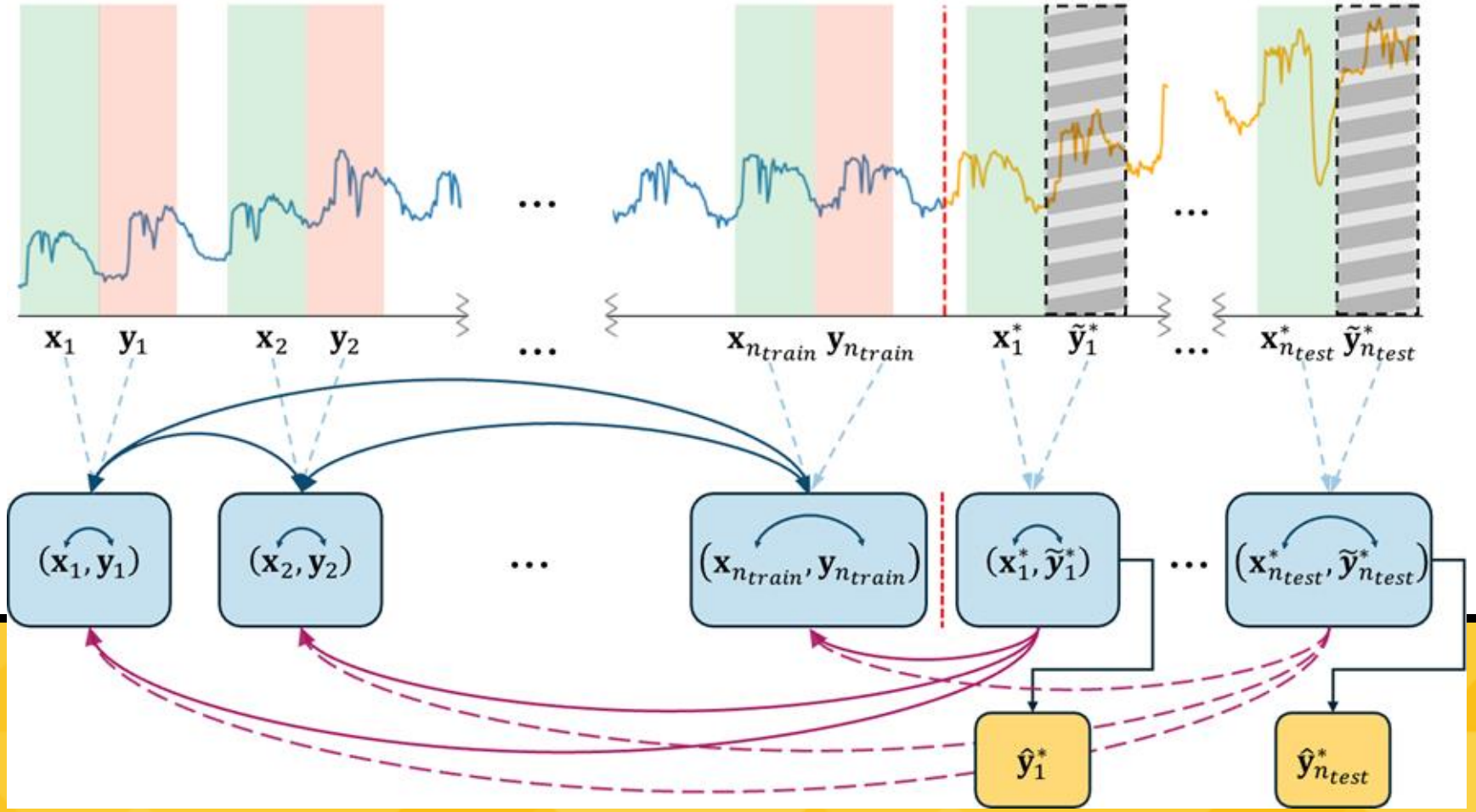


(b) Traditional Models



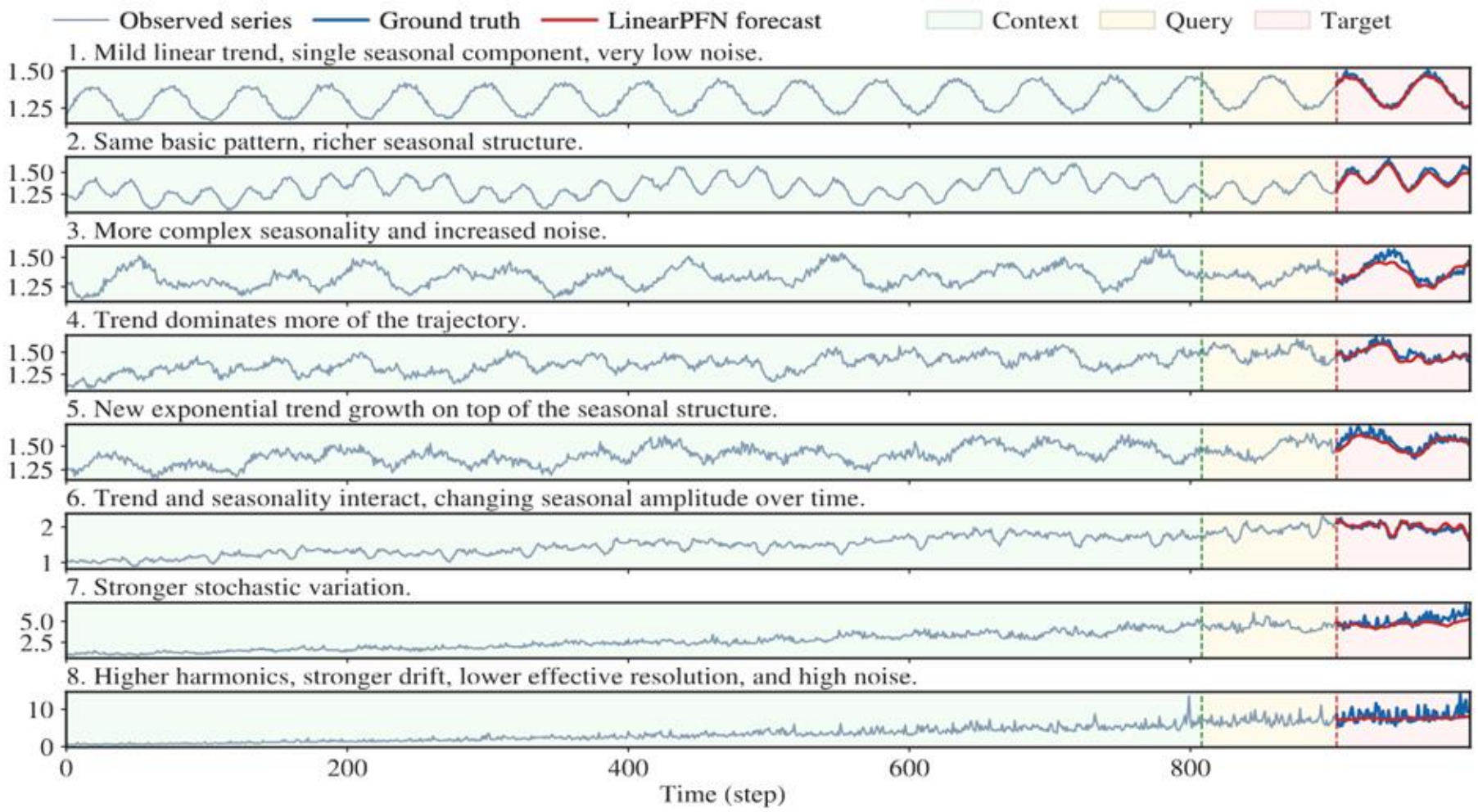
(c) Prior-data Fitted Networks

Difference between traditional time-series forecasting models and PFN models



Core architectural motivation

Results

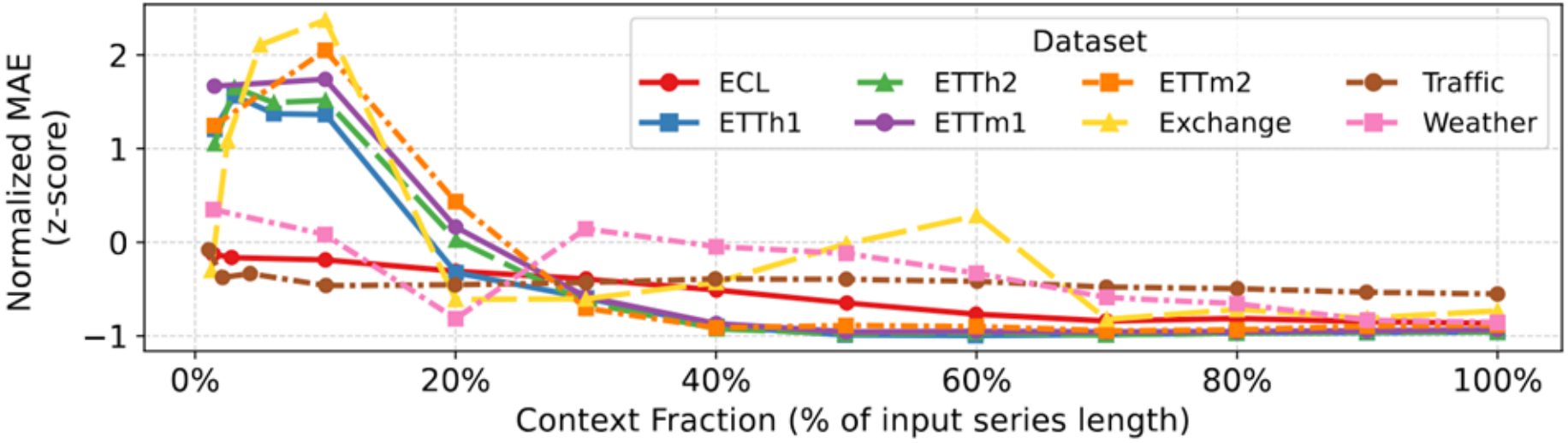

Observed series
Ground truth
LinearPFN forecast
Context
Query
Target
1. Mild linear trend, single seasonal component, very low noise.
2. Same basic pattern, richer seasonal structure.
3. More complex seasonality and increased noise.
4. Trend dominates more of the trajectory.
5. New exponential trend growth on top of the seasonal structure.
6. Trend and seasonality interact, changing seasonal amplitude over time.
7. Stronger stochastic variation.
8. Higher harmonics, stronger drift, lower effective resolution, and high noise.
1.50
1.25
2
1
5.0
2.5
10
0
0
200
400
600
800
Time (step)

Model	ECL		ETTh1		ETTh2		ETTm1		ETTm2		Exchange		Traffic		Weather	
	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE
Zero-shot																
LinearPFN	**0.387**	0.310	0.196	0.068	**0.275**	**0.128**	0.137	0.034	0.205	0.080	0.267	0.125	**0.344**	**0.249**	0.0308	0.015
TimePFN	0.590	0.613	0.186	0.060	0.294	0.143	0.139	0.034	0.251	0.110	0.272	0.126	0.507	0.435	0.0251	**0.012**
S. Naive	1.017	1.673	0.209	0.072	0.434	0.304	0.141	0.035	0.360	0.231	0.230	0.093	1.557	3.674	0.0257	0.013
Last	0.996	1.635	0.203	0.069	0.423	0.295	0.136	0.033	0.354	0.228	**0.221**	**0.088**	1.532	3.628	**0.0248**	**0.012**
Mean	0.794	0.965	0.198	0.067	0.359	0.206	0.149	0.038	0.305	0.152	0.302	0.147	1.173	1.862	0.0263	0.013
Supervised																
Informer	0.618	0.639	0.396	0.216	0.412	0.262	0.228	0.082	0.266	0.115	0.785	0.890	0.993	1.436	0.0843	0.106
FEDformer	0.461	0.374	0.216	0.077	0.326	0.171	0.163	0.043	0.225	0.085	0.307	0.157	0.529	0.459	0.0423	0.029
DLinear	0.609	0.611	0.215	0.082	0.345	0.189	0.154	0.041	0.245	0.101	0.310	0.142	0.781	0.887	0.0592	0.057
Crossformer	0.437	0.349	0.328	0.157	0.350	0.190	0.193	0.060	0.223	0.085	0.524	0.412	0.581	0.598	0.0427	0.030
Autoformer	0.569	0.560	0.217	0.077	0.305	0.155	0.180	0.052	0.333	0.155	0.314	0.157	0.489	0.425	0.0777	0.091
PatchTST	0.439	0.363	**0.183**	**0.058**	0.308	0.156	**0.127**	**0.029**	**0.196**	**0.072**	0.235	0.101	0.534	0.499	0.0293	0.016
iTransformer	0.399	**0.295**	0.195	0.065	0.327	0.174	0.132	0.031	0.216	0.082	0.257	0.119	0.426	0.354	0.0260	0.013

Comparison with traditional forecasting models

Model	ECL		ETTh1		ETTh2		Exchange		Weather		Traffic	
	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE
LinearPFN-36	**0.429**	**0.357**	0.146	0.039	**0.213**	**0.081**	0.170	0.045	0.2732	0.105	**0.400**	**0.303**
TimePFN-36	0.703	0.752	0.130	**0.029**	0.273	0.126	0.125	0.027	0.1381	0.042	1.032	1.503
ForecastPFN	0.958	1.416	0.237	0.102	0.517	0.434	0.180	0.057	0.1999	0.084	1.742	4.521
Chronos-s	0.792	1.152	0.155	0.061	0.321	0.207	0.113	0.049	0.1136	0.036	1.364	3.103
S. Naive	0.995	1.559	0.151	0.039	0.408	0.279	0.128	0.028	0.1352	0.045	1.771	4.301
Last	0.829	1.211	**0.128**	0.031	0.336	0.215	**0.107**	**0.022**	**0.1123**	**0.035**	1.463	3.330
Mean	0.805	0.963	0.154	0.040	0.321	0.168	0.156	0.040	0.1893	0.069	1.256	2.125

Zero-shot results on univariate time-series forecasting



Effect of context budget on forecasting error across datasets

Summary & Future Work

Summary & Future Work

- **Approach**: Adapts Prior-Data Fitted Networks (PFNs) to time-series forecasting, treating forecasting as in-context inference rather than learning a fixed input–output mapping
- **Key finding**: Strong zero-shot performance without large-scale pretraining, using only a small synthetic corpus
- **Architecture**: A patch-based embedder with a single linear embedding layer captures both short- and long-term dependencies
- **Training**: Meta-trained on synthetic data built from trend, seasonality, and noise components
- **Results**: Competitive performance on 8 benchmarks with a much simpler architecture than transformer-based baselines
- **Limitation and future work**: The current prior is bounded. Richer priors, such as GP-based (TimePFN) or TabPFN-style sampling, are promising but hard to adapt to patch structure
- **Takeaway**: PFN-style in-context learning is a viable, scalable, and interpretable foundation for forecasting

Thank You!